IoT Protocol Simulation - LO4 Lab Screenshots

Course: ITAI 3377
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This document contains evidence screenshots for the LO4 IIoT Protocol Simulation lab, organized by protocol and step.

Section 1: Environment Setup Steps 1-5 - Project folder, virtual environment, package installation, and Mosquitto broker

```
PS C:\Users\brand> python --version
Python 3.10.9
PS C:\Users\brand> mkdir iiot_simulation


    Directory: C:\Users\brand


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         2/14/2026   1:10 PM                iiot_simulation


PS C:\Users\brand> cd iiot_simulation
PS C:\Users\brand\iiot_simulation>
```

Step 1 – Project Folder Created and Opened

```
PS C:\Users\brand> cd iiot_simulation
PS C:\Users\brand\iiot_simulation> python -m venv venv
```

Step 2 – Virtual Environment Created

```
PS C:\Users\brand\iiot_simulation> python -m venv venv
PS C:\Users\brand\iiot_simulation> venv\Scripts\activate
(venv) PS C:\Users\brand\iiot_simulation>
```

Step 3 – Virtual Environment Activated

```
(venv) PS C:\Users\brand\iiot_simulation> pip install pandas numpy paho-mqtt
 aiocoap asyncua matplotlib
Collecting pandas
  Downloading pandas-2.3.3-cp310-cp310-win_amd64.whl (11.3 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 11.3/11.3 MB 3.1 MB/s eta 0:00:00
Collecting numpy
  Using cached numpy-2.2.6-cp310-cp310-win_amd64.whl (12.9 MB)
Collecting paho-mqtt
  Downloading paho_mqtt-2.1.0-py3-none-any.whl (67 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 67.2/67.2 kB 3.6 MB/s eta 0:00:00
Collecting aiocoap
  Downloading aiocoap-0.4.17-py3-none-any.whl (247 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 247.4/247.4 kB 3.0 MB/s eta 0:00:00
Collecting asyncua
  Downloading asyncua-1.1.8-py3-none-any.whl (1.1 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.1/1.1 MB 3.0 MB/s eta 0:00:00
Collecting matplotlib
  Downloading matplotlib-3.10.8-cp310-cp310-win_amd64.whl (8.1 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 8.1/8.1 MB 2.7 MB/s eta 0:00:00
Collecting tzdata>=2022.7
  Downloading tzdata-2025.3-py2.py3-none-any.whl (348 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 348.5/348.5 kB 4.3 MB/s eta 0:00:00
```

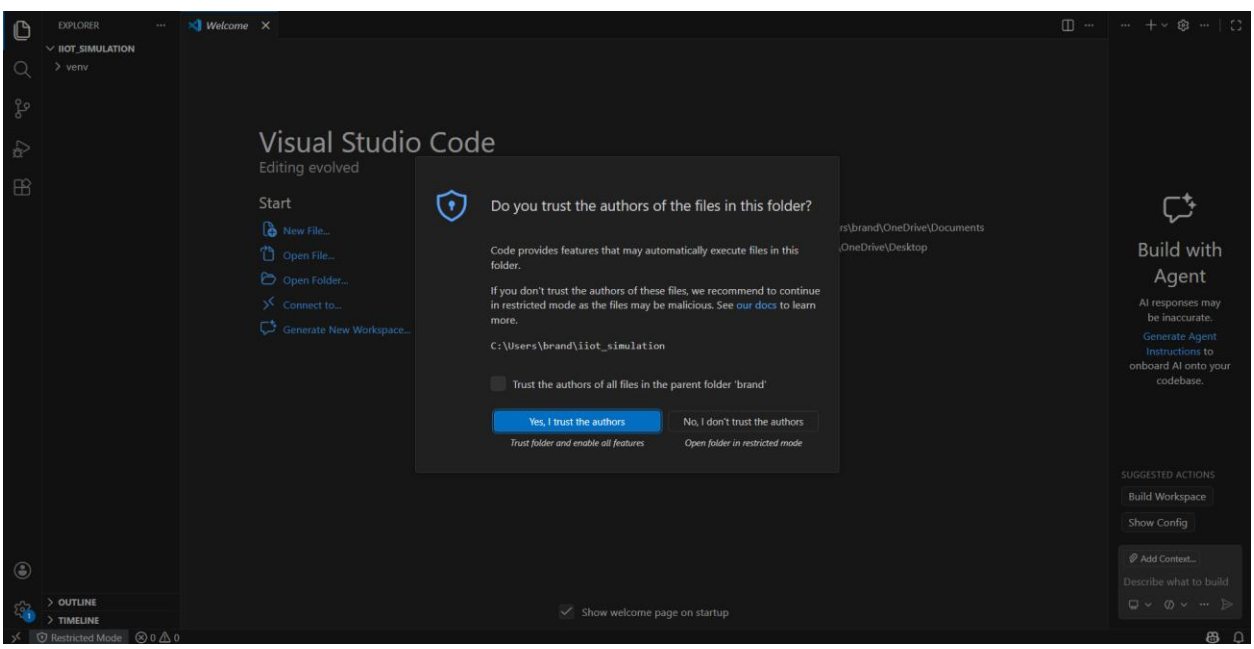
```
Installing collected packages: wait-for2, sortedcontainers, pytz, tzdata, ty
ping-extensions, six, pyparsing, pycparser, pillow, paho-mqtt, packaging, nu
mpy, kiwisolver, fonttools, cycler, aiosqlite, aiofiles, aiocoap, python-dat
eutil, contourpy, cffi, pandas, matplotlib, cryptography, pyopenssl, asyncua
  DEPRECATION: wait-for2 is being installed using the legacy 'setup.py insta
ll' method, because it does not have a 'pyproject.toml' and the 'wheel' pack
age is not installed. pip 23.1 will enforce this behaviour change. A possibl
e replacement is to enable the '--use-pep517' option. Discussion can be foun
d at https://github.com/pypa/pip/issues/8559
  Running setup.py install for wait-for2 ... done
```

```
  Running setup.py install for wait-for2 ... done
Successfully installed aiocoap-0.4.17 aiofiles-25.1.0 aiosqlite-0.22.1 async
ua-1.1.8 cffi-2.0.0 contourpy-1.3.2 cryptography-46.0.5 cycler-0.12.1 fontto
ols-4.61.1 kiwisolver-1.4.9 matplotlib-3.10.8 numpy-2.2.6 packaging-26.0 pah
o-mqtt-2.1.0 pandas-2.3.3 pillow-12.1.1 pycparser-3.0 pyopenssl-25.3.0 pypar
sing-3.3.2 python-dateutil-2.9.0.post0 pytz-2025.2 six-1.17.0 sortedcontaine
rs-2.4.0 typing-extensions-4.15.0 tzdata-2025.3 wait-for2-0.3.2

[notice] A new release of pip available: 22.3.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(venv) PS C:\Users\brand\iiot_simulation>
```

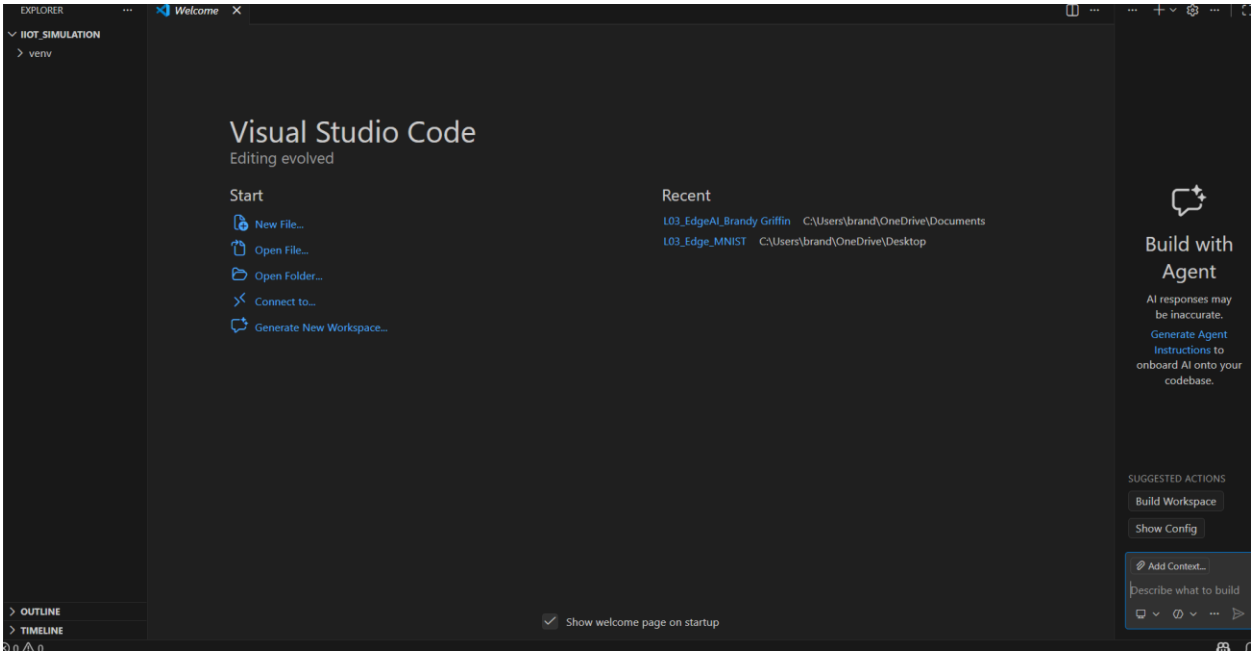
Step 4 – Required Python Packages Installed

```
(venv) PS C:\Program Files\mosquitto> .\mosquitto.exe
(venv) PS C:\Program Files\mosquitto> Get-Service mosquitto

Status   Name               DisplayName
------   ----               -----------
Running  mosquitto          Mosquitto Broker


(venv) PS C:\Program Files\mosquitto>
```

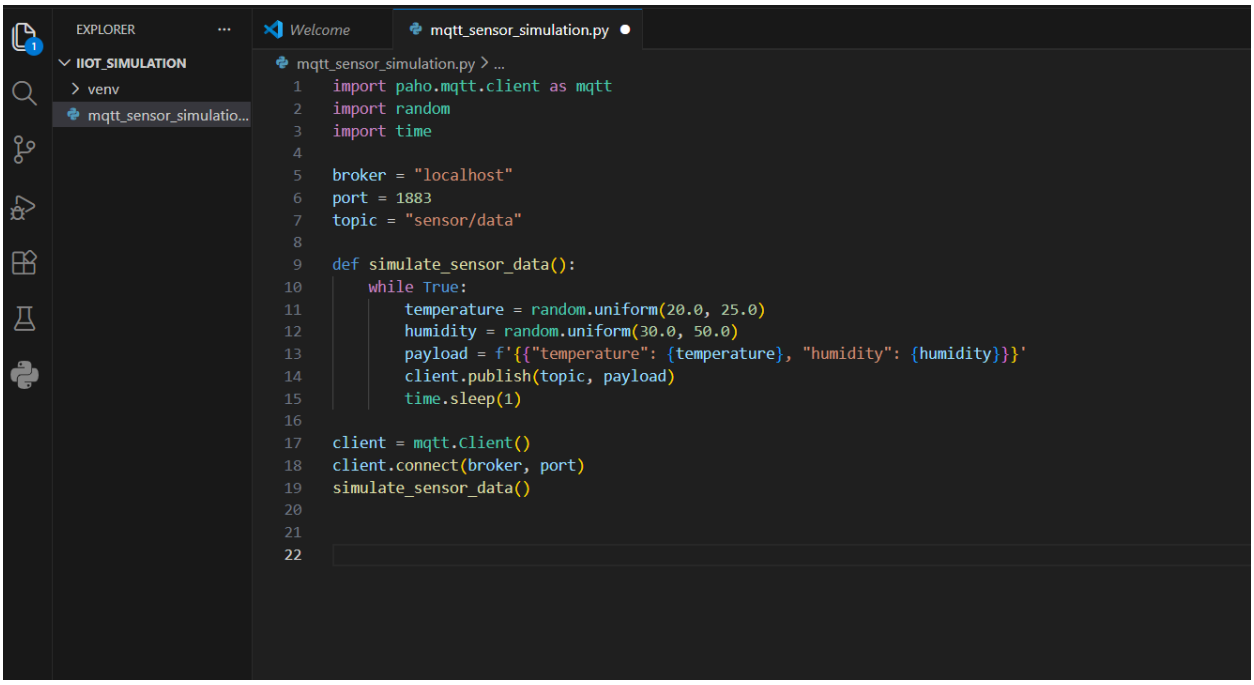
Step 5 – Mosquitto Broker Running as Windows Service



Step – VS Code Folder Trust Confirmed



Step – MQTT Script Created in Correct Folder



Step – MQTT Script Created in Correct Folder

```
    Directory: C:\Users\brand\iiot_simulation


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         2/14/2026   1:17 PM                venv
-a----         2/14/2026   1:49 PM            490 mqtt_sensor_simulation.py


(venv) PS C:\Users\brand\iiot_simulation>
```

```
(venv) PS C:\Users\brand\iiot_simulation> python mqtt_sensor_simulation.py
C:\Users\brand\iiot_simulation\mqtt_sensor_simulation.py:17: DeprecationWarn
ing: Callback API version 1 is deprecated, update to latest version
  client = mqtt.Client()
```

Step 6 – MQTT Sensor Simulation Running (With Deprecation Warning)

Section 2: MQTT Simulation Steps 6-8 - Sensor script, visualization script, runtime error, and live data output

```
(venv) PS C:\Users\brand\iiot_simulation> code .
(venv) PS C:\Users\brand\iiot_simulation> dir


    Directory: C:\Users\brand\iiot_simulation


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         2/14/2026   1:17 PM                venv
-a----         2/14/2026   2:02 PM           1049 data_visualization.py
-a----         2/14/2026   1:49 PM            490 mqtt_sensor_simulation.py


(venv) PS C:\Users\brand\iiot_simulation>
```

Step 7 – Visualization Script Created

```
    self.loop_forever(retry_first_connection=True)
  File "C:\Users\brand\iiot_simulation\venv\lib\site-packages\paho\mqtt\clie
nt.py", line 2297, in loop_forever
    rc = self._loop(timeout)
  File "C:\Users\brand\iiot_simulation\venv\lib\site-packages\paho\mqtt\clie
nt.py", line 1686, in _loop
    rc = self.loop_read()
  File "C:\Users\brand\iiot_simulation\venv\lib\site-packages\paho\mqtt\clie
nt.py", line 2100, in loop_read
    rc = self._packet_read()
  File "C:\Users\brand\iiot_simulation\venv\lib\site-packages\paho\mqtt\clie
nt.py", line 3142, in _packet_read
    rc = self._packet_handle()
  File "C:\Users\brand\iiot_simulation\venv\lib\site-packages\paho\mqtt\clie
nt.py", line 3808, in _packet_handle
    return self._handle_publish()
  File "C:\Users\brand\iiot_simulation\venv\lib\site-packages\paho\mqtt\clie
nt.py", line 4145, in _handle_publish
    self._handle_on_message(message)
  File "C:\Users\brand\iiot_simulation\venv\lib\site-packages\paho\mqtt\clie
nt.py", line 4501, in _handle_on_message
    on_message(self, self._userdata, message)
  File "C:\Users\brand\iiot_simulation\data_visualization.py", line 27, in o
n_message
    plt.pause(0.1)
  File "C:\Users\brand\iiot_simulation\venv\lib\site-packages\matplotlib\pyp
lot.py", line 757, in pause
    canvas.start_event_loop(interval)
  File "C:\Users\brand\iiot_simulation\venv\lib\site-packages\matplotlib\bac
kends\_backend_tk.py", line 475, in start_event_loop
    self._tkcanvas.mainloop()
  File "C:\Users\brand\AppData\Local\Programs\Python\Python310\lib\tkinter\_
_init__.py", line 1458, in mainloop
    self.tk.mainloop(n)
RuntimeError: Calling Tcl from different apartment
```

Step 8 – MQTT Visualization Runtime Error

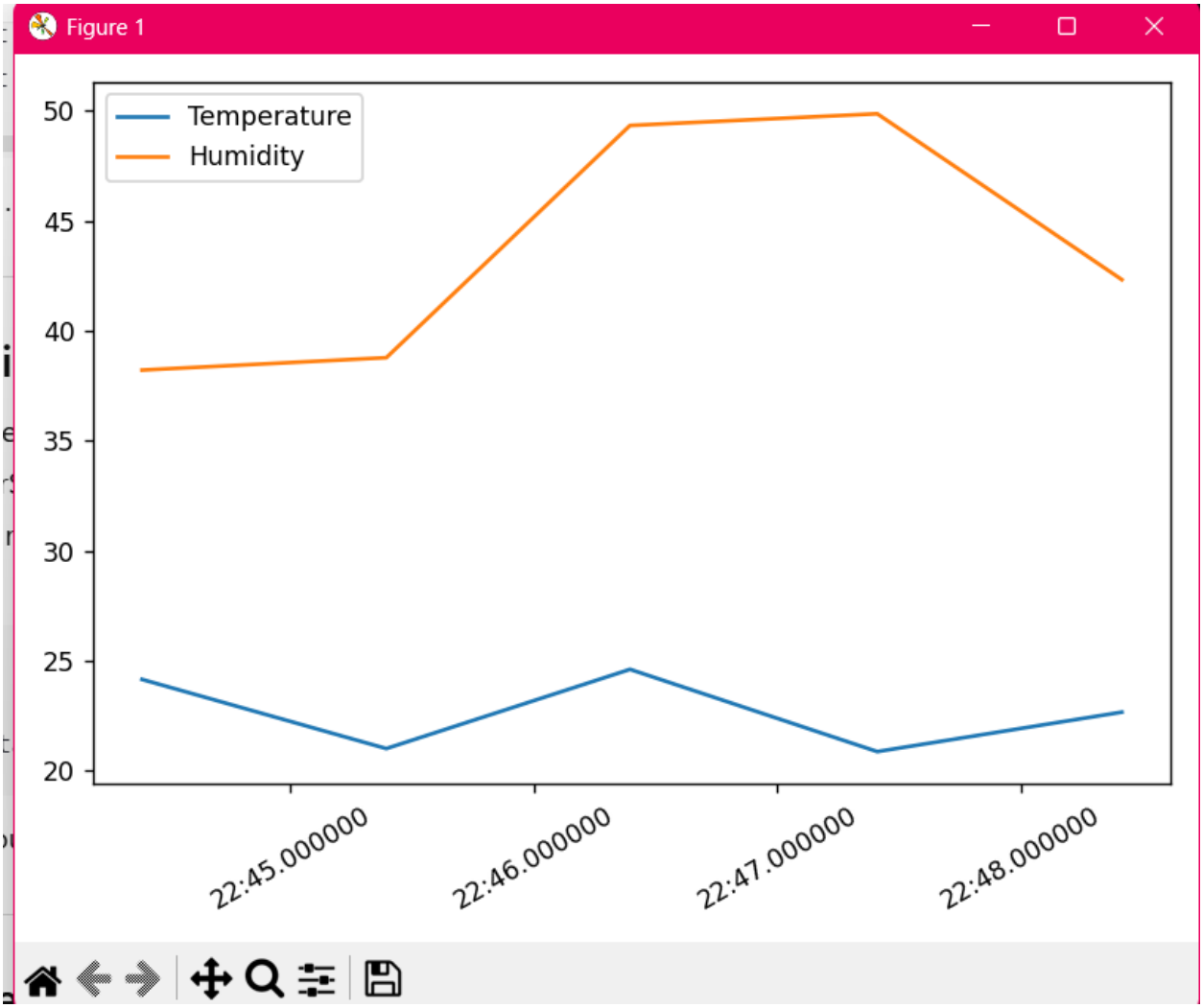


Figure 1
Temperature
Humidity
50
45
40
35
30
25
20
22:45.000000
22:46.000000
22:47.000000
22:48.000000

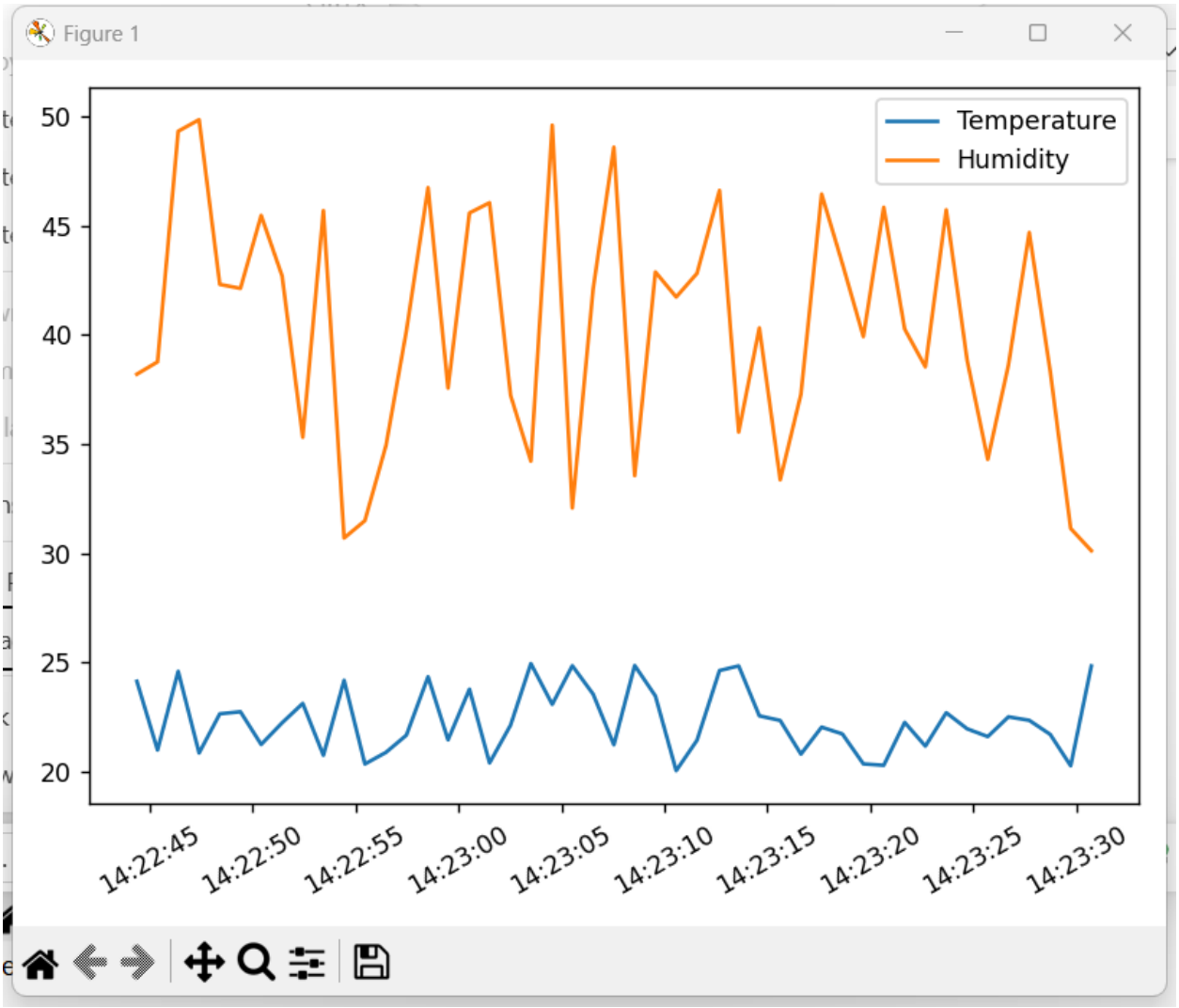
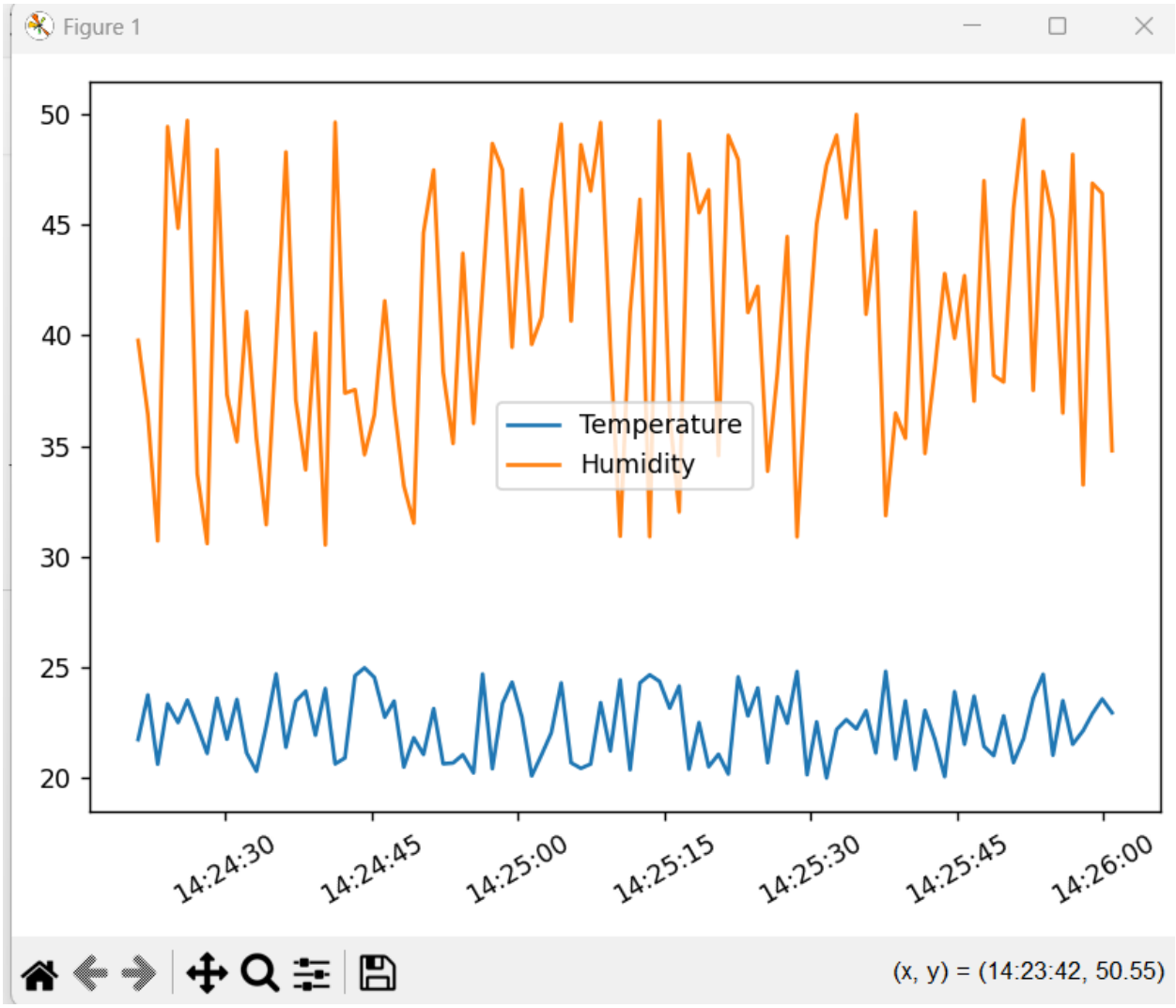
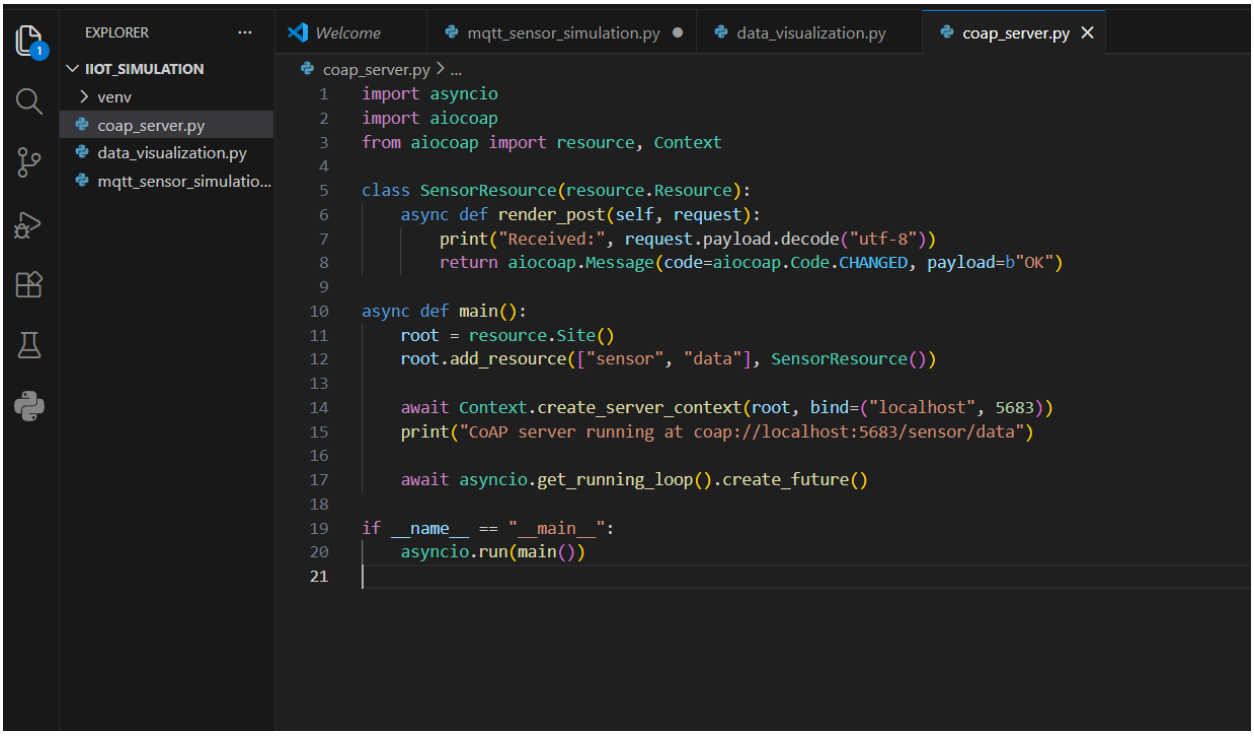


Figure 1
Temperature
Humidity
50
45
40
35
30
25
20
14:22:45
14:22:50
14:22:55
14:23:00
14:23:05
14:23:10
14:23:15
14:23:20
14:23:25
14:23:30



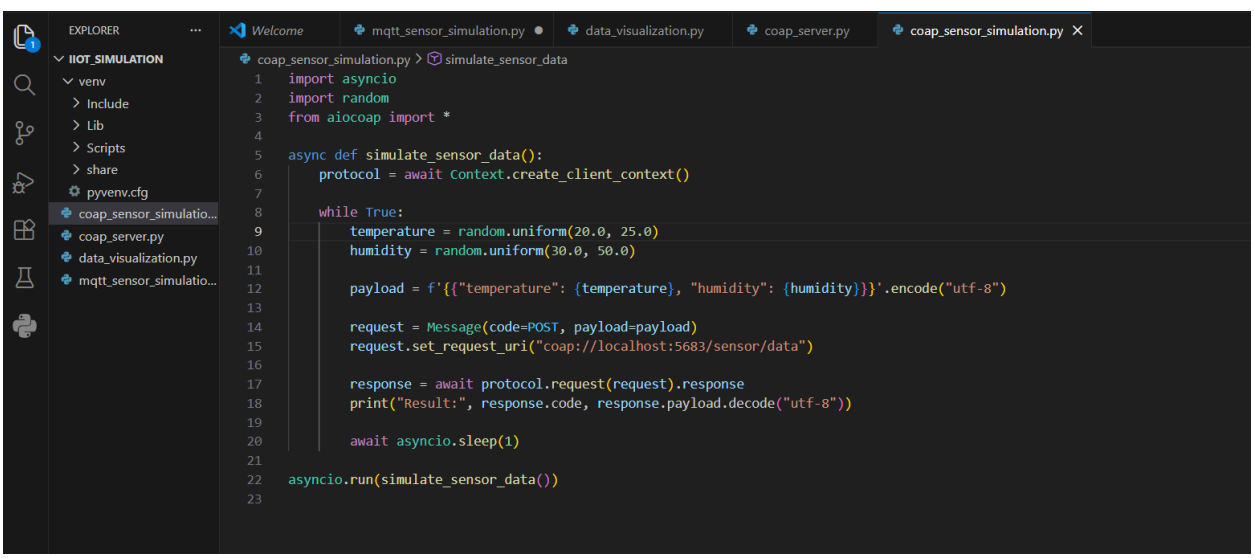
Step 8 – Live MQTT Data Visualization Working

Section 3: CoAP Simulation Steps C1-C6 — Server and client scripts, directory confirmation, running the stack, and troubleshooting



Screenshot – CoAP Step 1

"Step C1 – coap_server.py Created"



Screenshot – CoAP Step 2

"Step C2 – coap_sensor_simulation.py Created"

```
(venv) PS C:\Users\brand\iiot_simulation> cd C:\Users\brand\iiot_simulation
(venv) PS C:\Users\brand\iiot_simulation> dir


    Directory: C:\Users\brand\iiot_simulation


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         2/14/2026   1:17 PM                venv
-a----         2/14/2026   2:30 PM            696 coap_sensor_simulation.py
-a----         2/14/2026   2:27 PM            660 coap_server.py
-a----         2/14/2026   2:22 PM           1288 data_visualization.py
-a----         2/14/2026   1:49 PM            490 mqtt_sensor_simulation.py


(venv) PS C:\Users\brand\iiot_simulation>
```

Screenshot – CoAP Step 3

"Step C3 – CoAP Files Confirmed in Directory"

```
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Install the latest PowerShell for new features and improvements! https://aka
.ms/PSWindows

PS C:\Users\brand> cd C:\Users\brand\iiot_simulation
PS C:\Users\brand\iiot_simulation> venv\Scripts\activate
(venv) PS C:\Users\brand\iiot_simulation> python coap_server.py
CoAP server running at coap://localhost:5683/sensor/data
```

Screenshot – CoAP Step 4

"Step C4 – CoAP Server Running"

```
Install the latest PowerShell for new features and improvements! https://aka.ms/PSW
indows

PS C:\Users\brand> cd C:\Users\brand\iiot_simulation
PS C:\Users\brand\iiot_simulation> venv\Scripts\activate
(venv) PS C:\Users\brand\iiot_simulation> python coap_sensor_simulation.py
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
```

Screenshot – CoAP Step 5

Step C5 – CoAP Client Sending Data (Result: 2.04 Changed OK)

```
rt) is normally permitted
(venv) PS C:\Users\brand\iiot_simulation> netstat -ano | findstr :5683
  TCP    127.0.0.1:5683         0.0.0.0:0              LISTENING       21880
  TCP    [::1]:5683             [::]:0                 LISTENING       21880
  UDP    [::]:59098             [::1]:5683                             2524
  UDP    [::1]:5683             *:*                                    21880
(venv) PS C:\Users\brand\iiot_simulation>
```

CoAP Troubleshooting – Port 5683 In Use

```
rt) is normally permitted
(venv) PS C:\Users\brand\iiot_simulation> netstat -ano | findstr :5683
  TCP    127.0.0.1:5683         0.0.0.0:0              LISTENING       21880
  TCP    [::1]:5683             [::]:0                 LISTENING       21880
  UDP    [::]:59098             [::1]:5683                             2524
  UDP    [::1]:5683             *:*                                    21880
(venv) PS C:\Users\brand\iiot_simulation> taskkill /PID 21880 /F
>>
SUCCESS: The process with PID 21880 has been terminated.
(venv) PS C:\Users\brand\iiot_simulation>
```

CoAP Troubleshooting – Killed Existing Process on Port 5683

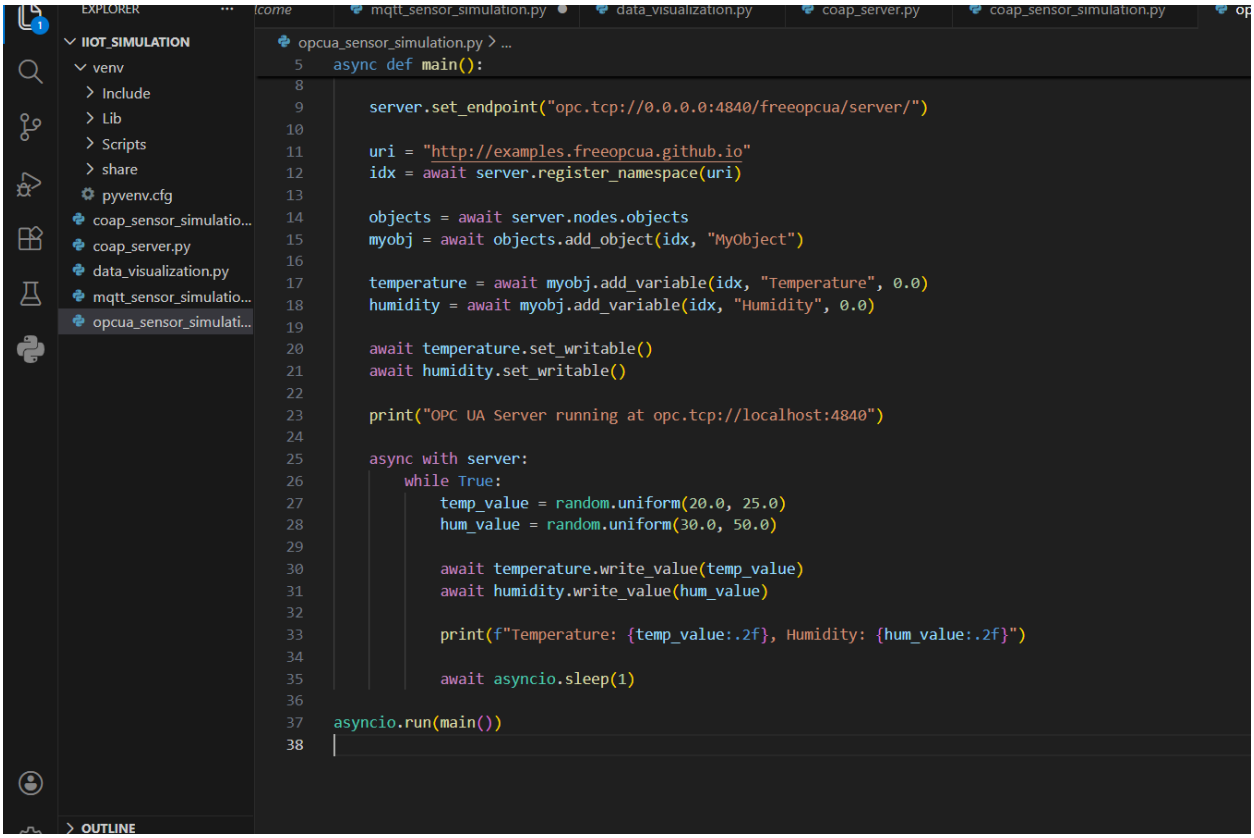
```
PS C:\Users\brand> python coap_sensor_simulation.py
C:\Users\brand\AppData\Local\Programs\Python\Python310\python.exe: can't ope
n file 'C:\\Users\\brand\\coap_sensor_simulation.py': [Errno 2] No such file
 or directory
PS C:\Users\brand> cd C:\Users\brand\iiot_simulation
PS C:\Users\brand\iiot_simulation> venv\Scripts\activate
(venv) PS C:\Users\brand\iiot_simulation> python coap_sensor_simulation.py
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
Result: 2.04 Changed OK
```

Step C5 – CoAP Client Successfully Sending Data (2.04 Changed OK)

```
Received: {"temperature": 24.800985678781096, "humidity": 39.99152552333835}
Received: {"temperature": 23.701971224443522, "humidity": 47.08816272022624}
Received: {"temperature": 22.788775855046516, "humidity": 31.916390996487067
}
Received: {"temperature": 24.983979755762142, "humidity": 32.657510204671816
}
Received: {"temperature": 22.852295802379384, "humidity": 38.55305212457}
Received: {"temperature": 22.43522812606118, "humidity": 37.40104036033877}
Received: {"temperature": 20.41669337078346, "humidity": 33.22551873497125}
Received: {"temperature": 24.406724488740608, "humidity": 41.11434633929395}
Received: {"temperature": 23.678943010161532, "humidity": 40.50474343125237}
Received: {"temperature": 22.93513453208233, "humidity": 45.35170809588467}
Received: {"temperature": 20.63507037672905, "humidity": 48.17949450911991}
Received: {"temperature": 24.909410597088456, "humidity": 43.74286640141501}
Received: {"temperature": 22.59780813556712, "humidity": 44.27715893683519}
Received: {"temperature": 21.14358893296672, "humidity": 46.576961787340466}
Received: {"temperature": 21.622358688608802, "humidity": 30.977000537828484
}
Received: {"temperature": 22.741404919520527, "humidity": 33.748279649801745
}
```

Step C6 – CoAP Server Receiving Data

Section 4: OPC UA Simulation Steps 01-03 — Script creation, directory confirmation, and live sensor data output



Step O1 – OPC UA Script Created

```
KeyboardInterrupt
(venv) PS C:\Users\brand\iiot_simulation> cd C:\Users\brand\iiot_simulation
(venv) PS C:\Users\brand\iiot_simulation> venv\Scripts\activate
(venv) PS C:\Users\brand\iiot_simulation> dir


    Directory: C:\Users\brand\iiot_simulation


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         2/14/2026   1:17 PM                venv
-a----         2/14/2026   2:30 PM            696 coap_sensor_simulation.py
-a----         2/14/2026   2:27 PM            660 coap_server.py
-a----         2/14/2026   2:22 PM           1288 data_visualization.py
-a----         2/14/2026   1:49 PM            490 mqtt_sensor_simulation.py
-a----         2/14/2026   3:00 PM           1073 opcua_sensor_simulation.p
                                                  y


(venv) PS C:\Users\brand\iiot_simulation>
```

Step O2 – OPC UA File Confirmed in Directory

```
s\brand\iiot_simulation\opcua_sensor_simulation.py", line 14,

wait server.nodes.objects
ct Node can't be used in 'await' expression
ers\brand\iiot_simulation> code .
ers\brand\iiot_simulation> python opcua_sensor_simulation.py
unning at opc.tcp://localhost:4840
olicy available, password may get transferred in plaintext
 than open requested but private key and certificate are not
```

```
Temperature: 21.85, Humidity: 44.85
Temperature: 24.00, Humidity: 30.46
Temperature: 23.18, Humidity: 36.96
Temperature: 20.50, Humidity: 36.39
Temperature: 23.79, Humidity: 30.63
Temperature: 22.92, Humidity: 39.67
Temperature: 21.19, Humidity: 32.88
Temperature: 20.80, Humidity: 43.86
Temperature: 22.15, Humidity: 41.98
Temperature: 21.38, Humidity: 41.26
Temperature: 23.61, Humidity: 46.38
Temperature: 22.77, Humidity: 43.50
Temperature: 22.39, Humidity: 45.12
Temperature: 21.78, Humidity: 49.00
Temperature: 21.43, Humidity: 48.80
Temperature: 23.48, Humidity: 37.17
Temperature: 23.99, Humidity: 34.83
Temperature: 23.83, Humidity: 41.01
Temperature: 24.01, Humidity: 38.33
Temperature: 22.42, Humidity: 40.40
Temperature: 22.56, Humidity: 47.75
Temperature: 24.54, Humidity: 36.57
Temperature: 22.65, Humidity: 46.62
Temperature: 20.24, Humidity: 46.70
Temperature: 22.75, Humidity: 33.23
Temperature: 24.59, Humidity: 48.53
Temperature: 22.98, Humidity: 46.11
Temperature: 20.38, Humidity: 35.43
Temperature: 20.69, Humidity: 44.18
Temperature: 23.48, Humidity: 36.87
Temperature: 24.85, Humidity: 34.86
Temperature: 22.23, Humidity: 37.88
Temperature: 20.95, Humidity: 39.50
```

Step O3 – OPC UA Server Generating Sensor Data in Real Time

```
PS C:\Users\brand\iiot_simulation> cd C:\Users\brand\iiot_simulation
PS C:\Users\brand\iiot_simulation> dir


    Directory: C:\Users\brand\iiot_simulation


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         2/14/2026   1:17 PM                venv
d-----         2/15/2026   4:59 PM                visualizations
-a----         2/15/2026  10:59 AM          85445 coap_data.csv
-a----         2/15/2026   4:27 PM            929 coap_sensor_simulat
                                                  ion.py
-a----         2/15/2026   4:27 PM           1623 coap_server.py
-a----         2/15/2026   4:28 PM           2588 coap_visualization.
                                                  py
-a----         2/15/2026   1:48 PM           2033 data_visualization.
                                                  py
-a----         2/15/2026   1:36 PM            928 mqtt_sensor_simulat
                                                  ion.py
-a----         2/14/2026   3:00 PM           1073 opcua_sensor_simula
                                                  tion.py


PS C:\Users\brand\iiot_simulation>
```

Conclusion:

This lab brought three core IIoT protocols - MQTT, CoAP, and OPC UA - to life in a single local environment, showing not just how each one works in theory but what it actually takes to get them running, troubleshoot them when they break, and understand why you would choose one over another for a given use case. Every protocol threw at least one real problem along the way, whether it was a threading conflict in the MQTT visualizer, a port collision that had to be cleared manually before CoAP could start, or an async/await mistake in the OPC UA script that required going back into the code and understanding the library more carefully before it would run. What stands out most is how different the three protocols feel in practice - MQTT is quick and forgiving, CoAP is clean and RESTful but demands more attention to the network layer, and OPC UA carries real weight in both setup and capability, making it clear why industrial systems reach for it when reliability and structured data matter more than simplicity. Taken together, these simulations build a solid foundation for working with real IIoT hardware, where the same protocols appear but with far less room for error.